

Module 4 – Assignment Questions

1. What is the significance of MAC layer and in which position it is placed in the OSI model

Significance of the MAC Layer:

- **Channel Access Control:** Determines when and how devices access the shared wireless medium using **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance).
- **Frame Formatting:** Defines the structure of 802.11 frames including addressing, control fields, and data.
- **Addressing:** Uses MAC addresses (48-bit) to identify source and destination devices on the network.
- **Reliability:** Handles acknowledgment (**ACK**) of frames and retransmission of lost frames.
- **Association and Authentication:** Manages the process of clients joining the **BSS** (Basic Service Set).
- **Power Management:** Controls when client devices can enter sleep mode and when the AP buffers frames for them.
- **Security:** Implements encryption (WPA2/WPA3) and authentication (802.1X/EAPOL) at the MAC level.
- **QoS:** Prioritizes traffic using 802.11e (WMM) Access Categories for voice, video, best-effort, and background traffic.
- **Fragmentation and Aggregation:** Splits large frames into smaller fragments or combines small frames for efficiency.

2. Describe the frame format of the 802.11 MAC header and explain the purpose of each fields

Field	Size	Purpose
Frame Control	2 bytes	Defines the frame type (Management/Control/Data), subtype, and various flags (To DS, From DS, More Frag, Retry, Power Mgmt, More Data, Protected, Order)
Duration/ID	2 bytes	Specifies the time (in microseconds) the channel will be occupied — used by other devices to update their NAV (Network Allocation Vector) for virtual carrier sensing
Address 1 (RA)	6 bytes	Receiver Address — the immediate recipient of the frame (next hop). Always present.
Address 2 (TA)	6 bytes	Transmitter Address — the device that is transmitting the frame. Always present.
Address 3	6 bytes	Depends on To DS / From DS bits. Could be BSSID, source, or destination address depending on frame direction.
Sequence Control	2 bytes	Contains Fragment Number (4 bits) and Sequence Number (12 bits) — used to reassemble fragmented frames and discard duplicates
Address 4	6 bytes	Used only in WDS (Wireless Distribution System) or mesh — carries the original source address in 4-address frames
QoS Control	2 bytes	Present in QoS Data frames — specifies the Traffic Identifier (TID) / Access Category for prioritization
HT Control	4 bytes	Present in HT/VHT frames — carries link adaptation and beamforming feedback information
Frame Body	0–7951 bytes	The actual payload — management info, control data, or upper-layer data (e.g., IP packets)
FCS	4 bytes	Frame Check Sequence — CRC-32 checksum to detect transmission errors

3. Please list all the MAC layer functionalities in all Management, Control and Data plane

Management Plane – Frame Types and Functions:

Management frames handle the establishment and maintenance of communication between devices. They are never encrypted in legacy 802.11 (PMF – Protected Management Frames in 802.11w adds protection).

Frame Subtype	Function
Beacon	AP periodically broadcasts its presence, SSID, supported rates, and capabilities
Probe Request	Client broadcasts to find available APs (active scanning)
Probe Response	AP responds to a Probe Request with its capabilities
Authentication	First step of joining — Open System or SAE authentication exchange
Association Request	Client requests to join the BSS and sends its capabilities
Association Response	AP accepts or rejects the association request
Reassociation Request/Response	Used during roaming to move association from one AP to another
Disassociation	Terminates the association — can be sent by AP or client
Deauthentication	Terminates the authentication — forces client to restart join process
Action	Carries specific management actions (spectrum management, QoS, block ACK setup, beamforming)

Control Plane – Frame Types and Functions:

Control frames assist in the delivery of data and management frames. They ensure reliable delivery and manage channel access.

Frame Subtype	Function
ACK (Acknowledgment)	Confirms successful receipt of a unicast frame — sent by receiver after every unicast data/management frame
RTS (Request to Send)	Sent before a large data frame to reserve the channel and avoid hidden node collisions
CTS (Clear to Send)	Sent in response to RTS — confirms channel is clear; also used as CTS-to-self for protection
Block ACK Request (BAR)	Requests a block acknowledgment for a set of frames sent in an aggregated burst
Block ACK (BA)	Acknowledges multiple frames at once using a bitmap — one ACK for up to 64 frames

Data Plane – Frame Types and Functions:

Data frames carry the actual user payload (IP packets, etc.) between devices.

Frame Subtype	Function
Data	Basic data frame carrying upper-layer payload
Null Data	Carries no payload — used by clients to notify AP of power save state change (no data, just Power Management bit)
QoS Data	Data frame with QoS Control field — carries Traffic Identifier for WMM prioritization
QoS Null	QoS version of Null frame — used for power management signaling with QoS-capable APs

4. Explain the scanning process and its types in detail

The **scanning process** is the mechanism by which a Wi-Fi client (station) discovers the Access Points (APs) available in its vicinity. This is the first step a device takes before it can authenticate and associate with a network.

There are two primary methods for scanning: **Passive Scanning** and **Active Scanning**.

1. Passive Scanning

In passive scanning, the client simply "listens" to each channel to hear what APs are broadcasting. It does not transmit any frames.

- **Mechanism:** The client moves to a channel and waits for a **Beacon Frame**. Every AP sends out a Beacon at a regular interval (typically every 100 milliseconds) to announce its presence, SSID, supported rates, and security capabilities.
- **Time:** The client must stay on each channel long enough to ensure it doesn't miss the Beacon (usually slightly more than 100ms per channel).
- **Pros:** Saves battery because the radio isn't transmitting; it is also "quiet," creating less congestion on the network.
- **Cons:** It is slower than active scanning because the client is at the mercy of the AP's Beacon interval.

2. Active Scanning

In active scanning, the client takes a "hands-on" approach by specifically asking what networks are available.

- **Mechanism:**
 1. The client moves to a channel and sends a **Probe Request** frame. This request can be "broadcast" (asking for any SSID) or "targeted" (asking for a specific hidden SSID).
 2. Any AP on that channel that matches the request will respond with a **Probe Response** frame, which contains almost the same information as a Beacon.
- **Pros:** Much faster discovery. The client doesn't have to wait 100ms; it sends the probe, waits a few milliseconds for a response, and moves to the next channel.

- **Cons:** Higher battery consumption; increases "management overhead" traffic on the airwaves.

3. The Scanning Workflow

Regardless of the type, the client follows a specific sequence:

1. **Channel Switching:** The client tunes its radio to a specific frequency (e.g., Channel 1).
2. **Listening/Probing:** It either waits for a Beacon (Passive) or sends a Probe Request (Active).
3. **Data Collection:** It records the signal strength (RSSI), SSID, and capabilities of every AP it hears.
4. **Repeat:** It moves to the next channel (e.g., Channel 6) and repeats the process until all supported frequencies (2.4 GHz and 5 GHz) have been scanned.
5. **Selection:** Once the scan is complete, the client uses its internal logic (often picking the strongest signal or a "known" SSID) to start the **Authentication** process.

5. Brief about the client association process

The **client association process** is the multi-step handshake that a device (Station/STA) must complete to join a Basic Service Set (BSS) managed by an Access Point (AP). Once a client has finished the **Scanning Process** we discussed earlier and has selected a target AP, it must move through two logical states: **Authentication** and **Association**.

Step 1: 802.11 Authentication (The "Low-Level" Handshake)

Before a device can even request to join the network, it must prove it is a Wi-Fi device. Note that this is **not** where you enter your WPA2 password; this is a legacy connectivity requirement.

1. **Authentication Request:** The client sends a management frame to the AP asking to be authenticated.
2. **Authentication Response:** The AP sends a response back.
 - In modern networks, this usually uses **Open System Authentication**, which essentially means the AP says "Yes" to everyone at this stage.

Step 2: Association (The "Logical Connection")

Once authenticated, the client officially requests to become a member of the AP's network. This is where the specific "capabilities" of the connection are negotiated.

1. **Association Request:** The client sends its details to the AP, including:
 - Supported Data Rates (e.g., MCS indices).
 - Channel Width capabilities (e.g., 20MHz vs 40MHz).
 - Security capabilities (e.g., WPA2 or WPA3).

- Power Save capabilities.
- 2. **Association Response:** The AP reviews the request. If the AP's settings match the client's capabilities, it sends a "Success" message and assigns an **Association ID (AID)**. The AID is a unique number that identifies that specific device on the AP's map.

Step 3: Post-Association (Security and IP)

Technically, the 802.11 association is now complete, but the client cannot send internet data yet. Two final hurdles remain:

- **802.1X/EAP (If Enterprise):** If it's an office/university network, the **RADIUS** server handshake occurs now.
 - **4-Way Handshake:** The client and AP derive encryption keys.
 - **DHCP:** The client requests an **IP Address**. Only after an IP is assigned is the client truly "connected" to the internet.
6. Explain each steps involved in EAPOL 4-way handshake and the purpose of each keys derived from the process

The EAPOL (Extensible Authentication Protocol Over LAN) 4-Way Handshake is performed after 802.11 association to mutually verify knowledge of the PMK (Pairwise Master Key) and derive session encryption keys. It is used in WPA2-Personal and WPA2-Enterprise.

Key Hierarchy (Before the Handshake):

- **PMK (Pairwise Master Key):** In WPA2-Personal, PMK = PBKDF2(PSK, SSID, 4096 iterations). In WPA2-Enterprise, PMK is derived from EAP authentication with RADIUS server. The PMK is 256 bits and is the root of all session keys.
- **PTK (Pairwise Transient Key):** Derived during the handshake. Used for unicast frame encryption between AP and client.
- **GTK (Group Temporal Key):** Used to encrypt broadcast and multicast frames — shared among all clients on the BSS.

The 4-Way Handshake Steps:

Message 1 – AP → Client (ANonce):

- The AP sends its ANonce (AP Nonce — a random 256-bit number) to the client.
- Purpose: Provides the client with the AP's random value needed to derive the PTK.
- At this point, the client now has: PMK + ANonce + SNonce (its own) + AP MAC + Client MAC → derives the PTK.
- PTK derivation: $PTK = PRF(PMK, ANonce, SNonce, AP_MAC, Client_MAC)$

Message 2 – Client → AP (SNonce + MIC):

- The client sends its SNonce (Station Nonce) to the AP, along with a MIC (Message Integrity Code) computed using the KCK portion of the derived PTK.

- Purpose: Proves to the AP that the client has the correct PMK (it could only compute a valid MIC if it derived the correct PTK).
- The AP now also derives the PTK using the received SNonce.
- The AP verifies the MIC — if valid, the PMK is confirmed to match on both sides.

Message 3 – AP → Client (GTK + Install PTK):

- The AP sends the GTK (Group Temporal Key) encrypted with the PTK's KEK (Key Encryption Key).
- The AP also sets the Install flag, instructing the client to install the PTK.
- A MIC is included to prove integrity.
- Purpose: Delivers the group key securely and signals the client to start using the PTK for encryption.

Message 4 – Client → AP (Confirmation):

- The client sends a confirmation ACK to the AP with a MIC.
- Purpose: Confirms that the client has received the GTK and installed the keys.
- After this, both AP and client have installed the PTK and GTK — encrypted communication begins.

Key	Derived From	Size	Purpose
PMK	PSK + SSID (Personal) or EAP (Enterprise)	256 bits	Master key — root of all session keys; never transmitted over the air
PTK	PRF(PMK + ANonce + SNonce + MACs)	384–512 bits	Parent of KCK, KEK, and TK — used for unicast frame protection
KCK (Key Confirmation Key)	First 128 bits of PTK	128 bits	Computes the MIC in EAPOL messages 2, 3, 4 to verify integrity
KEK (Key Encryption Key)	Next 128 bits of PTK	128 bits	Encrypts the GTK in Message 3 so it is not sent in plaintext
GTK (Group Temporal Key)	Generated by AP	128/256 bits	Encrypts broadcast and multicast frames sent to all clients in the BSS

7. Describe the power saving scheme in MAC layer and explore on the types of Power saving mechanisms

A. Legacy Power Save (PS-Poll)

This is the original method where the client is in total control.

1. The client sees its bit is set in the Traffic Indication Map (TIM).
2. The client sends a **PS-Poll** (Power Save Poll) control frame to the AP.
3. The AP sends **one** buffered frame.

4. If the AP has more data, it sets a "More Data" bit in that frame, and the client sends another PS-Poll.
- **Drawback:** It is very "chatty" and inefficient for high-speed data.

B. WMM Power Save (U-APSD)

Introduced with 802.11e, **Unscheduled Automatic Power Save Delivery (U-APSD)** is much faster and better for applications like VoIP.

1. The client sends a "Trigger Frame" (any data frame).
2. The AP responds by sending a **burst** of multiple buffered frames in a single transmission.
3. The client doesn't have to poll for every single packet, which reduces overhead and saves more battery.

C. S-APSD (Scheduled APSD)

Similar to U-APSD, but the AP and client agree on a **fixed schedule** for when the client will wake up. This is highly efficient because the client doesn't even need to listen for Beacons; it just wakes up at the exact millisecond previously agreed upon.

D. Target Wake Time (TWT) — Wi-Fi 6 (802.11ax)

TWT is the most advanced power-saving feature. It allows an AP and a device (especially IoT sensors) to negotiate wake-up times that can be **hours or days apart**.

- **Benefit:** A smart water sensor can "sleep" for 24 hours, wake up for 10 milliseconds to report its status, and go back to sleep. This can extend battery life from months to years.

The "Power Management" Bit

How does the AP know a client is asleep? In the **802.11 MAC Header**, there is a **Frame Control** field containing a **Power Management bit**.

- If a client sends a frame (usually a **Null Data** frame) with this bit set to **1**, the AP knows the client is now in "Power Save" mode and will begin buffering its data.
- When the client is ready to stay awake, it sends a frame with the bit set to **0**.

8. Describe the Medium Access Control methodologies

Medium Access Control (MAC) methodologies define how devices share the wireless medium to avoid collisions and ensure fair access. Unlike wired Ethernet (CSMA/CD), wireless networks cannot detect collisions during transmission, so they use collision avoidance instead.

1. CSMA/CA – Carrier Sense Multiple Access with Collision Avoidance:

This is the primary MAC methodology in 802.11. Before transmitting, a device:

- **Listens to the medium (carrier sense)** — if busy, it waits.
- **Waits for DIFS (DCF Interframe Space)** after the medium becomes idle.

- **Selects a random backoff value** from a contention window and counts down.
- **Transmits only when the backoff counter reaches zero.**
- **The receiver sends an ACK** after successful receipt. If no ACK is received, the sender retransmits with a larger contention window (binary exponential backoff).

2. DCF – Distributed Coordination Function:

- The foundational MAC method — built on top of CSMA/CA.
- All devices compete equally for the medium using random backoff.
- Used for best-effort data traffic.
- No central coordination — fully distributed.

3. PCF – Point Coordination Function:

- Optional, centralized MAC method where the AP acts as a Point Coordinator.
- Uses a Contention-Free Period (CFP) where the AP polls clients one by one.
- Eliminates contention during CFP — deterministic access.
- Rarely implemented in practice — superseded by HCF.

4. HCF – Hybrid Coordination Function (802.11e):

HCF combines contention-based and contention-free access for QoS:

- **EDCA (Enhanced Distributed Channel Access):** Extends DCF with 4 Access Categories (AC_VO, AC_VI, AC_BE, AC_BK). Higher-priority ACs use shorter AIFS (Arbitration IFS) and smaller contention windows, giving them statistical priority.
- **HCCA (HCF Controlled Channel Access):** Centralized polling by AP for guaranteed service periods. Used for deterministic QoS. Rarely deployed.

5. Virtual Carrier Sense – NAV (Network Allocation Vector):

- Devices that hear a frame update their NAV timer based on the Duration field in the MAC header.
- They defer transmission until the NAV expires, even if they cannot hear the actual transmission.
- Used with RTS/CTS to solve the hidden node problem.

9. Brief about the Block ACK mechanism and its advantages

Setup Phase:

- The sender initiates a Block ACK session by sending an ADDBA (Add Block ACK) Request to the receiver.
- The request specifies: Buffer Size (max frames in a burst, e.g., 64), Starting Sequence Number, and TID (Traffic Identifier).
- The receiver responds with an ADDBA Response accepting or rejecting the request.

Transmission Phase:

- The sender transmits a burst of frames (up to the negotiated window size) without waiting for individual ACKs.
- Each frame has a sequence number.
- When done, the sender sends a BAR (Block ACK Request) frame specifying the starting sequence number.

Acknowledgment Phase:

- The receiver responds with a Block ACK (BA) frame containing a bitmap.
- Each bit in the bitmap corresponds to a frame sequence number — 1 = received, 0 = missing.
- The sender retransmits only the frames marked as missing (0 bits).
- The teardown is done via DELBA (Delete Block ACK) frames.

Types of Block ACK:

- Immediate Block ACK: BA is sent immediately after the BAR.
- Delayed Block ACK: BA may be sent after a delay (rarely used).
- Compressed Block ACK: Uses a 64-bit bitmap — acknowledges up to 64 frames (802.11n).
- Multi-TID Block ACK: Acknowledges frames across multiple Traffic Identifiers in one BA (802.11n).
- HE Block ACK (802.11ax): Extended bitmap for larger aggregation windows.

Advantages of Block ACK:

- Reduced Overhead: One Block ACK replaces up to 64 individual ACK frames — dramatically reduces the number of control frames on the medium.
- Higher Throughput: Less time spent on ACK transmissions = more time for data.
- Efficient with Aggregation: Block ACK is designed to work with A-MPDU — together they are the primary reason 802.11n/ac/ax can achieve hundreds of Mbps.
- Selective Retransmission: Only missing frames are retransmitted, not the entire burst.
- Lower Latency: Fewer round-trips between sender and receiver for acknowledgments.

10. Explain about A-MSDU, A-MPDU and A-MSDU in A-MPDU

□ **A-MSDU – Aggregate MAC Service Data Unit:**

- Aggregates multiple MSDUs (the data payloads from upper layers, e.g., IP packets) into a single MPDU (one MAC frame).
- Structure: Single MAC Header -> MSDU 1 | MSDU 2 | MSDU 3 | ... -> Single FCS.
- Each subframe has its own MSDU header (Destination Address, Source Address, Length).
- Maximum size: 7935 bytes (802.11n), 11,426 bytes (802.11ac/ax).
- All MSDUs must be for the same receiver and same TID (Traffic Identifier).
- Advantage: Reduces MAC header overhead — one MAC header for multiple payloads.
- Disadvantage: If any part of the A-MSDU is corrupted, the entire A-MSDU must be retransmitted (single FCS covers everything).
- No encryption boundary per MSDU — the entire A-MSDU is encrypted as one unit.

□ **A-MPDU – Aggregate MAC Protocol Data Unit:**

- Aggregates multiple MPDUs (complete MAC frames, each with its own header and FCS) into a single PHY layer transmission (PPDU).
- Structure: PLCP Header -> [MPDU Delimiter | MPDU 1 | MPDU Delimiter | MPDU 2 | ...] -> End.
- Each MPDU has its own MAC header, FCS, and is independently encrypted.
- Maximum number of MPDUs: 64 (802.11n), 256 (802.11ac/ax).
- Maximum A-MPDU size: 65,535 bytes (802.11n), 1,048,575 bytes (802.11ac).
- Works with Block ACK — the receiver can acknowledge which individual MPDUs were received using the bitmap.
- Advantage: If one MPDU is corrupted, only that MPDU needs retransmission — not the entire aggregate.
- More robust than A-MSDU for error-prone environments.

□ **A-MSDU in A-MPDU (Two-Level Aggregation):**

- This is the combination of both aggregation methods for maximum efficiency.
- Each MPDU in the A-MPDU can itself contain an A-MSDU — so multiple MSDUs are packed into each MPDU, and multiple such MPDUs are packed into one A-MPDU transmission.
- Structure: A-MPDU { [MPDU with A-MSDU 1] | [MPDU with A-MSDU 2] | ... }
- This provides: Reduced MAC header overhead (from A-MSDU) + Independent per-MPDU error recovery (from A-MPDU) + Maximum data density per PPDU.
- Used in 802.11n/ac/ax for maximum throughput scenarios.
- Limitation: Only applicable to frames that are not management frames and must share the same receiver and TID within each A-MSDU.

